Audited published FEM pilot

100 source cases audited | 3 admitted cases | 6 separate track tasks

The collection supports benchmark preparation, but it is not a certified benchmark of complex engineering capability.

Three original published calibration problems have sufficiently specified geometry, materials, loads, supports and FEM scalar references for input preparation. They are packaged in drawing-to-CAD-to-FEM and supplied-geometry-to-FEM tracks.

97 other collection cases are excluded or staged with explicit blockers. The five supplementary drawing documents were reviewed separately; they do not add a complete CAD/FEM case to the pilot.

189 distinct collection assets and 611 numerical reference values were verified against publisher files. Original solver inputs for the Bath lattices and meta-shells were acquired and inspected without executing them.

No engineering geometry, missing physics or reference FEM solution was generated. The physical inputs are source excerpts, with benchmark submission rules added around them. No frontier model trial was run.

Certification status: zero scored cases. Independent convergence/stress-recovery baselines, acceptance tolerances and an isolated replay runtime remain pending. Public examples may be present in model training data.

Admission decisions

Collection-wide audit; source assets remain preserved

SFEM (61)

Exclude deterministic solving

Stochastic medians; realized loads absent.

Bath curved links (18)

Stage complex geometry

Native IPT runtime and CAD/node mapping unverified.

Morphing-wing specimens (8)

Hold geometry mapping

CAD is 19 mm longer than the FEM domain.

Auxetic meta-shells (6)

Hold nonlinear mapping

Material fit, fixtures and torque/rotation mapping.

Scan-based columns (3)

Hold physics setup

Full material law and contact/loading unverified.

NAFEMS LE1/LE10/LE11 (3)

Admit calibration pilot

Source-complete inputs; no certified numerical pass.

NAFEMS GNL5 (1)

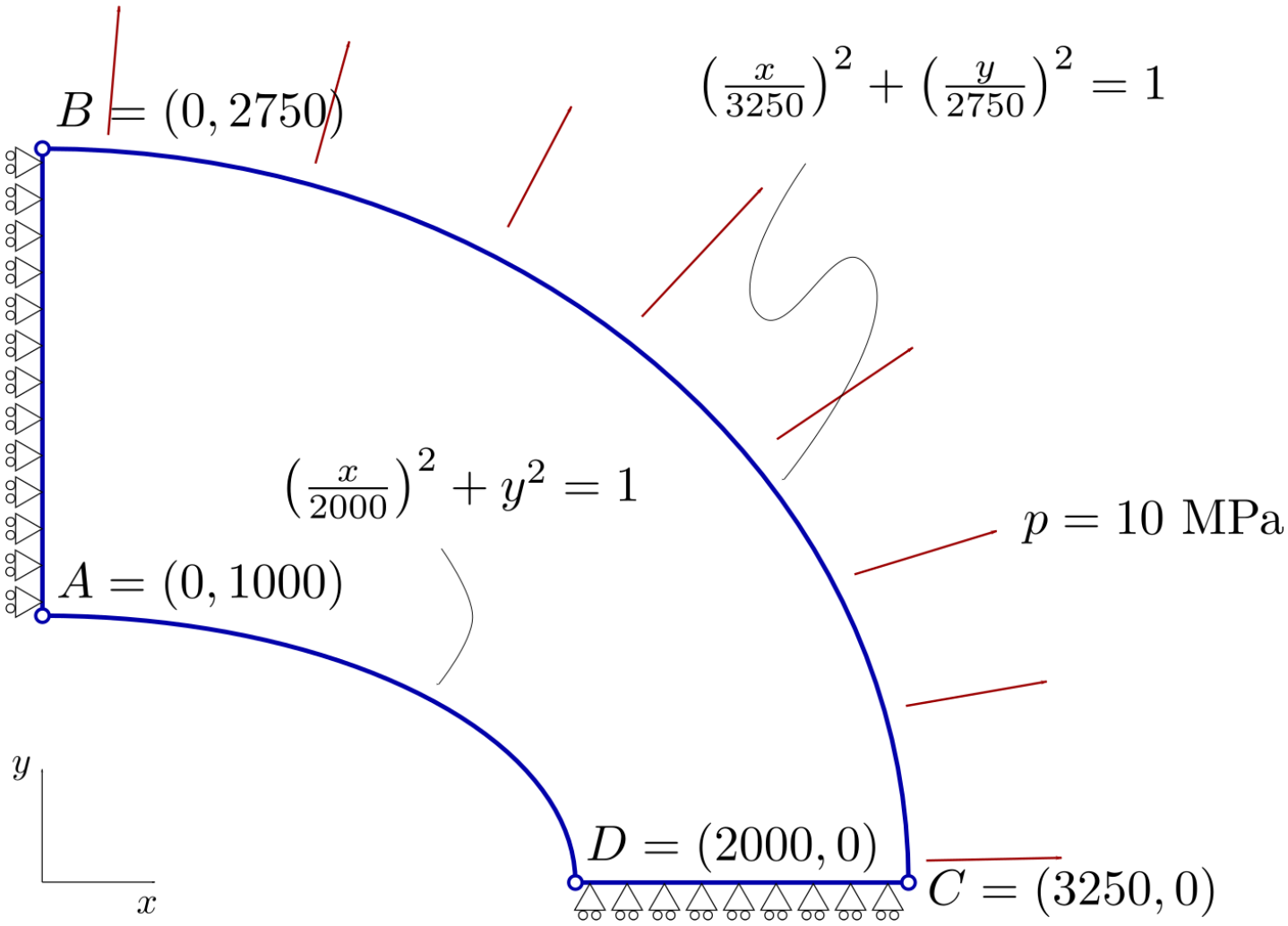
Exclude provisional reference

Source implementation is work in progress.

Supplement: the shaft and connecting-rod papers provide drawings and FEM values, but their complete physical specifications/decks remain unverified. Wheel and hoist sources have no FEM references.

NAFEMS-LE1: calibration case

Plane-stress elasticity | Source-complete inputs; scoring certification pending



Original published diagram. No new drawing or CAD is constructed in this report.

Formulation: 2D plane stress. The drawing inner-ellipse equation has a typo. Original parameters are a=1000, b=2750, c=3250, d=2000 mm; use the labelled endpoints.

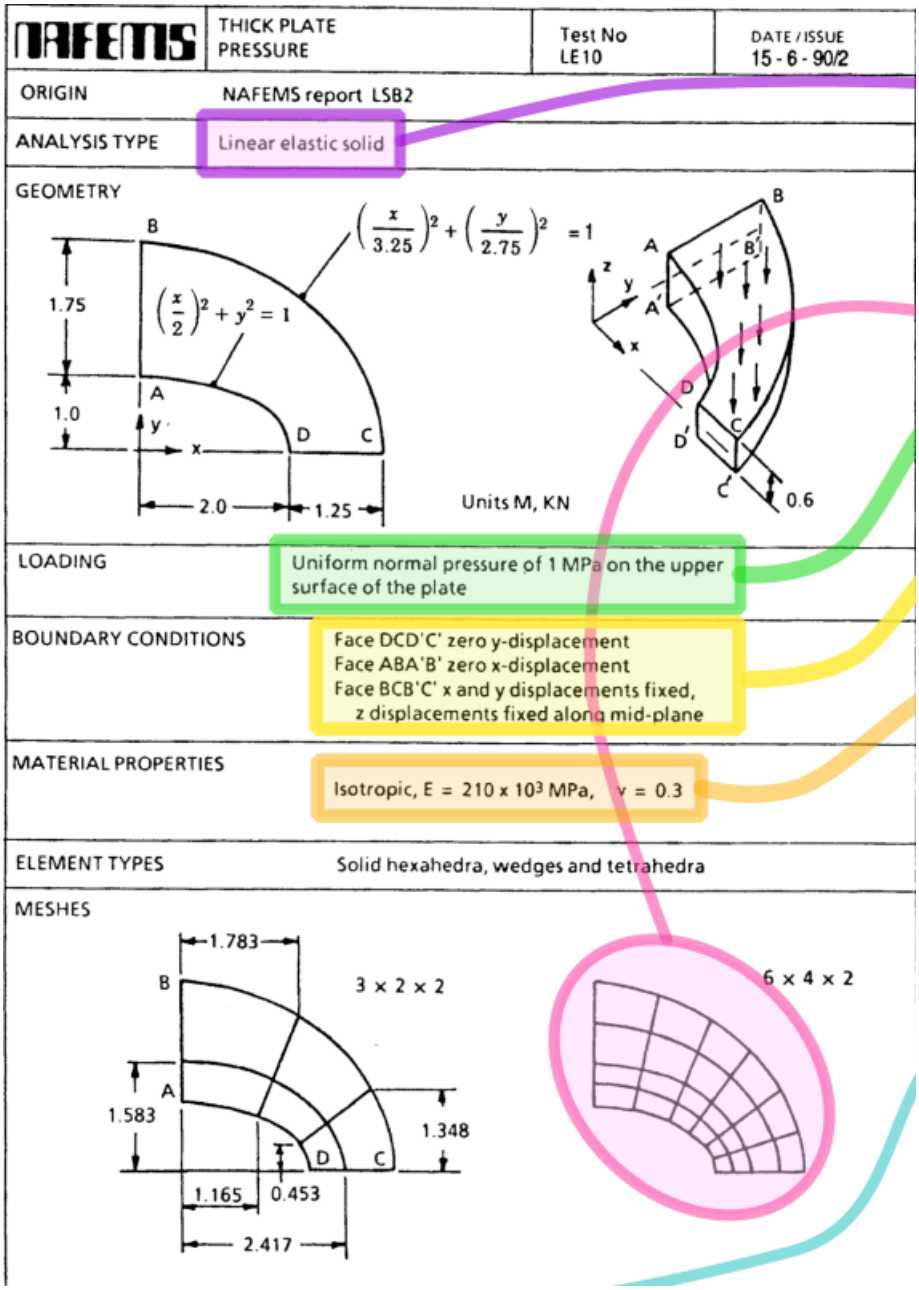
Held-out published FEM reference: sigma_yy_at_D = 92.7011 MPa. Query point: [2000, 0] mm.

Point-stress recovery and mesh refinement must be reviewed. Matching the number alone cannot establish a correct FEM solve.

Source: https://www.seamplex.com/feenox/examples/mechanical.html

NAFEMS-LE10: calibration case

Pressure-loaded elasticity | Source-complete inputs; scoring certification pending



Original published diagram. No new drawing or CAD is constructed in this report.

Formulation: 3D linear elasticity. Drawing dimensions are in metres; supplied source geometry uses mm. Pressure=1 MPa. All named boundary faces are defined in the drawing.

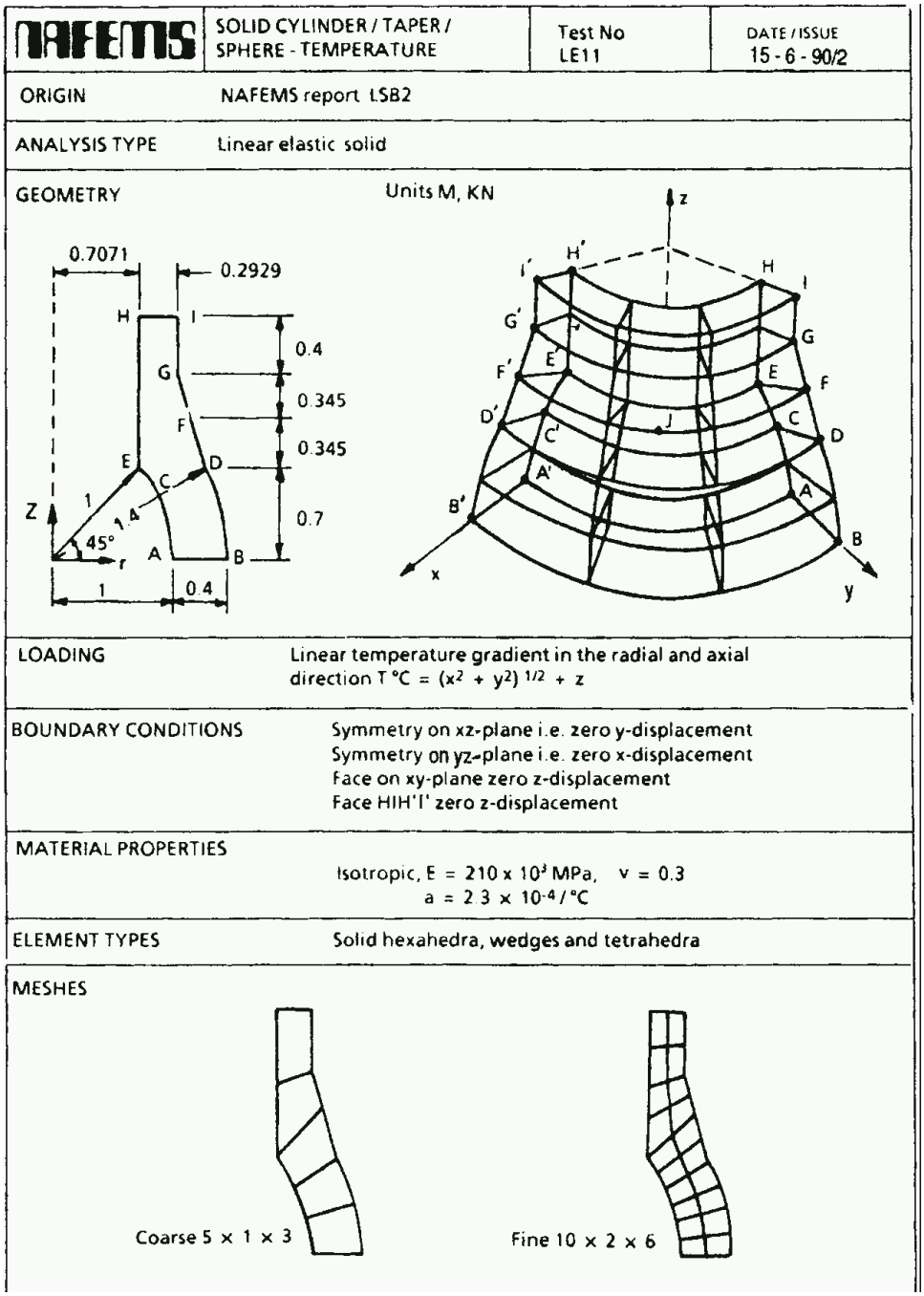
Held-out published FEM reference: sigma_yy_at_D = -5.37968 MPa. Query point: [2000, 0, 300] mm.

Point-stress recovery and mesh refinement must be reviewed. Matching the number alone cannot establish a correct FEM solve.

Source: https://www.seamplex.com/feenox/examples/mechanical.html

NAFEMS-LE11: calibration case

Thermoelasticity | Source-complete inputs; scoring certification pending



Original source drawing, with the answer panel removed from candidate inputs.

Formulation: 3D linear thermoelasticity. Geometry uses m.

Source uses the exact spherical junction z=sqrt(0.5), not the rounded 0.7 on the drawing. The two taper intervals are .345 m each. Thermal expansion is 2.3e-4 /degC, as published.

Stress-free reference temperature is 0 degC, taken from the pinned source solver default.

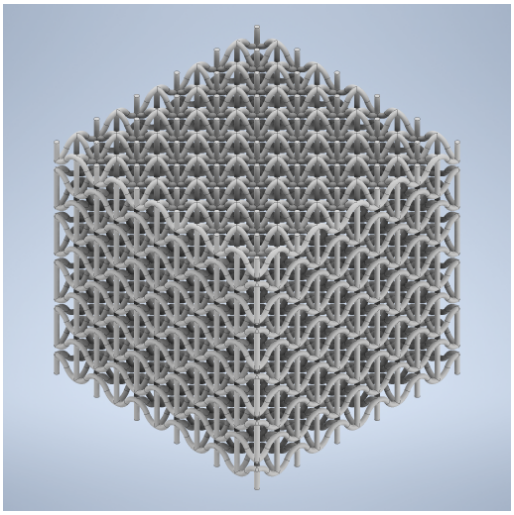
Held-out published FEM reference: sigma_zz_at_A = -105.04 MPa. Query point: [1, 0, 0] m.

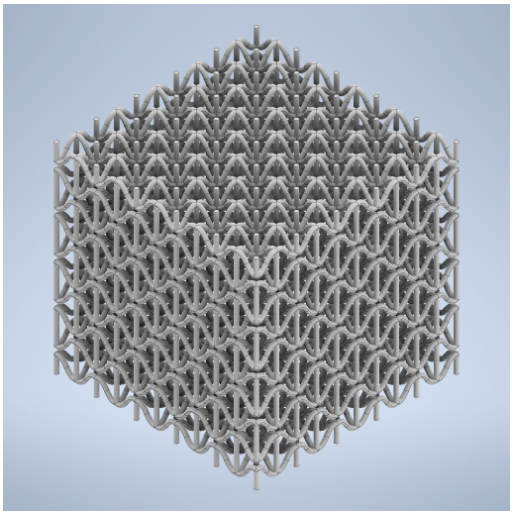
Point-stress recovery and mesh refinement must be reviewed. Matching the number alone cannot establish a correct FEM solve.

Source: https://www.seamplex.com/feenox/examples/mechanical.html

Complex candidates: source evidence

Original geometry/result assets exist; these cases remain outside the pilot





Bath curved lattices: original Inventor embedded previews. APDL decks contain MKS units, E=6.2 MPa, nu=0.3 and prescribed displacement loading with large-deformation analysis. These are source values, not new analysis results.

Admission still requires a usable native-CAD import route, a verified mapping from boundary node components to CAD surfaces, and agreement between the source deck and populated DP0 workbook outputs.

Auxetic meta-shells: the original archive has CW/CCW nonlinear self-contact decks and uniaxial material test data. The author visualization code rescales raw CSV torque to experiment. Raw FEM and calibrated experimental plots must remain distinct references.

Additional blockers: +/-6.283-rad deck rotation versus pi in the paper; missing export/frame/sign correspondence; fitted Ogden constants and integrated gripping-ring correspondence. The 31-degree CAD/CSV versus 30-degree input filename also remains unresolved.

Source: https://researchdata.bath.ac.uk/1525/

Source: https://zenodo.org/records/12800503

What model performance will mean

Two tracks; independent numerical and physical checks

Track A: drawing -> CAD -> FEM

Model receives the original diagram, source physical declarations and precision notes. It must construct CAD, generate/refine a mesh and solve the FEM problem. Reference CAD, solver recipes and numerical targets remain held out.

Track B: supplied geometry -> FEM

Model receives the same physical inputs plus original source geometry. It must implement the analysis and produce fields, a solver log and refinement history. Reports for the two tracks remain separate.

Numerical diagnostics

Report signed absolute error and relative error. Convert compatible units explicitly. Reject incompatible units, missing results and nonfinite values. No unsupported percentage threshold is assigned.

Physical validity

Trusted review checks topology/dimensions, formulation, materials, supports/loads, CAD-to-mesh correspondence, actual FEM execution and convergence/stress recovery. A copied matching scalar does not pass.

Runtime and contamination

Isolated replay has no network and no access to evaluator files. Docker is unavailable locally, so execution remains infrastructure-blocked. Public-source contamination remains possible; this pilot cannot establish generalisation.

Prepared artifacts and expansion

Preparation is complete; a broad certified capability benchmark is not

The six candidate packs contain source inputs without answers. A separate evaluator archive holds original assets and numerical contracts. Do not mix these archives in model context or candidate mounts.

The audit lists all 100 collection decisions and five supplementary document decisions, with hashes, line locators and blockers. The complex shortlist selects ten staged cases across lattices, self-contact shells, columns and a stepped modal specimen.

The runner freezes one-shot submissions, supports isolated replay, compares numerical outputs and requires evaluator-authored reviews bound to exact artifacts. It makes no model API requests.

Ten behavior tests cover admission, answer separation, unit conversion, signed errors, malformed results, submission paths, source-deck parsing and separate track reports. No tests are frontier-model scores.

Before a scored release: verify archived-problem baselines and stress recovery, derive supported tolerances, validate the isolated runtime and trusted review process, and resolve each complex case admission blocker.

Coverage gaps remain: manufacturing/forming, fatigue/fracture, transient dynamics, geared assemblies, complete pump/turbine assemblies and coupled fluid/thermal/structural tasks. No missing case or physical specification has been fabricated.